Zynq DIP2 LED 설계

Lab 요약

이 실습에서 Zynq PS에 PL Peripheral 를 연결하는 방법을 배운다.

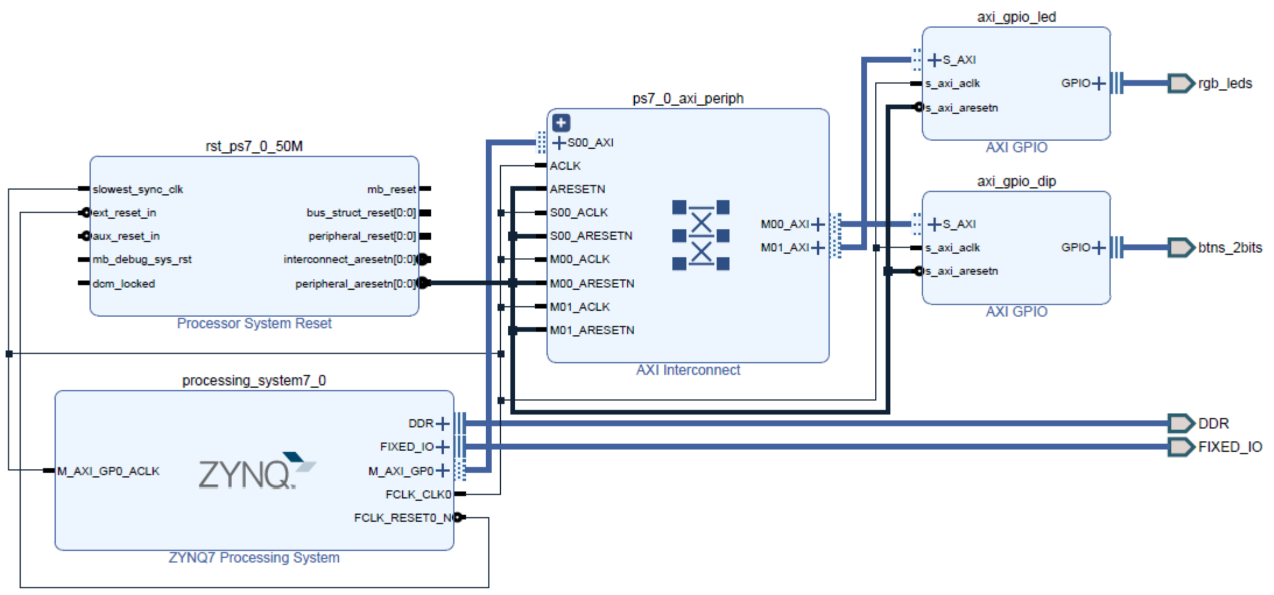
실습목적

- IP integrator 사용방법
- Zynq PS 설정방법 이해
- AXI interface 신호 확인
- PL IP(GPIO) 활용 방법
- Vitis를 이용한 PS Application 개발방법
- Zynq system 검증방법

개요

이 실습에에서는 Zynq system 을 구성하고 GPIO (dis switch, led) 를 PL에 추가하고 Vitis에서 동작을 확인한다.

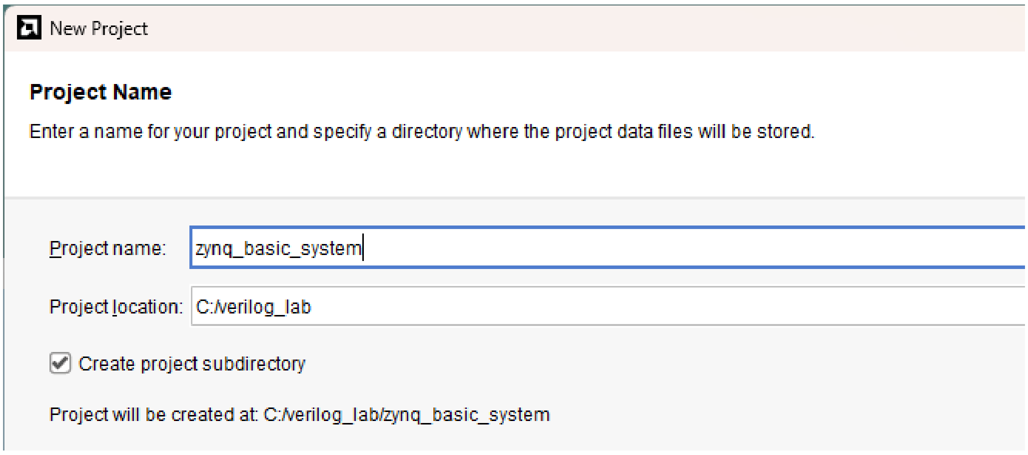
Zynq Basic System 시스템

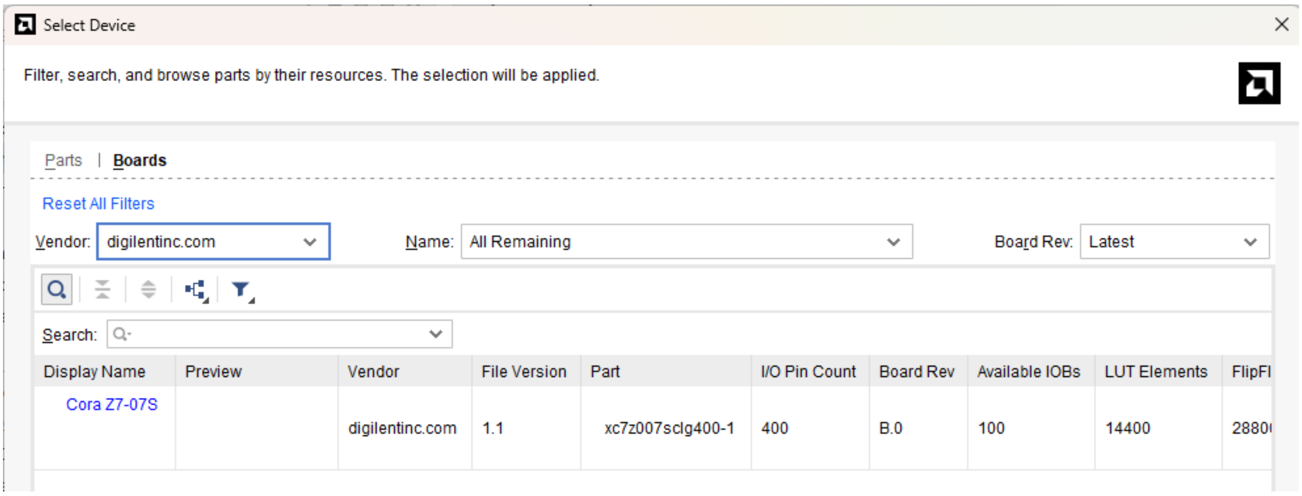


Step 1: Creating a New Vivado Design Suite Project → Step 2: Creating block design → Step 3: Zynq, gpio dip, gpio led 추가 → Step 4: synthesis & open synthesized design → Step 5: generate bitstream

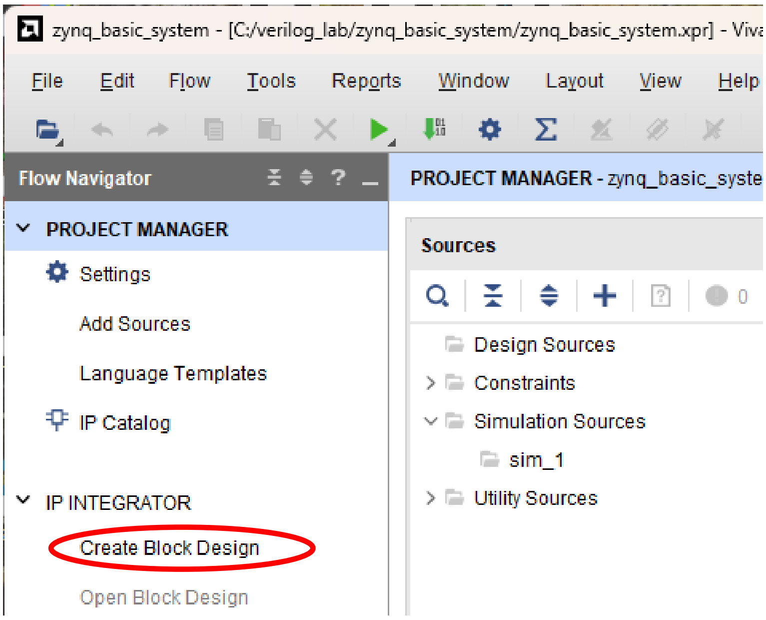
Step 6: Open hardware manager

1.1 Zynq basic system 프로젝트 생성

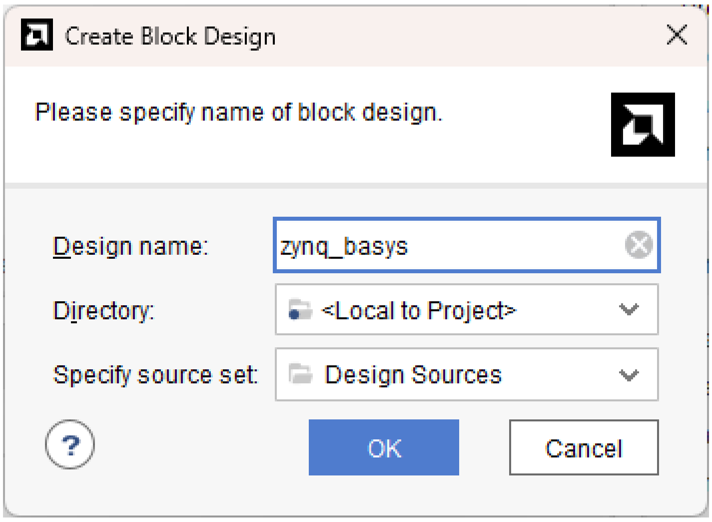




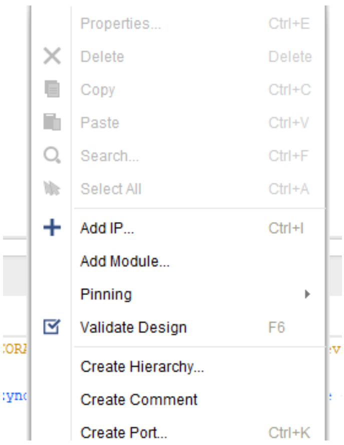
1.2 Create block design



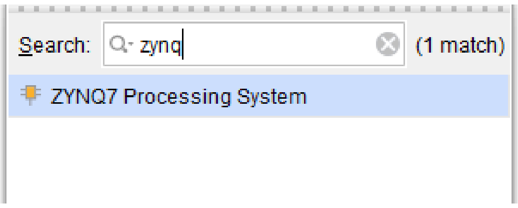
1.2.1 Flow navigator -> Project manager -> IP Integrator -> create block design 클릭



1.2.2 Design name 에 zynq_basys 입력 후 ok 클릭



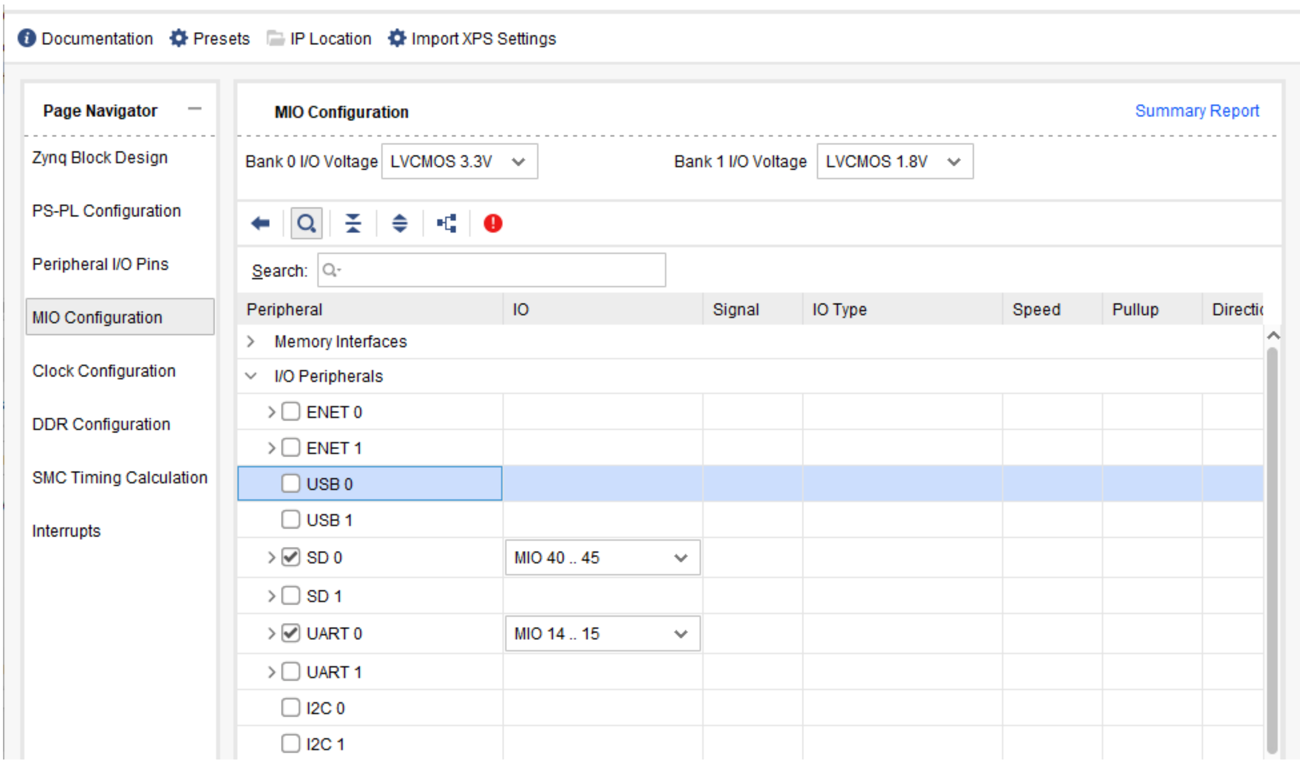
1.2.3 마우스 오른쪽 버튼 클릭 후 Add IP 클릭



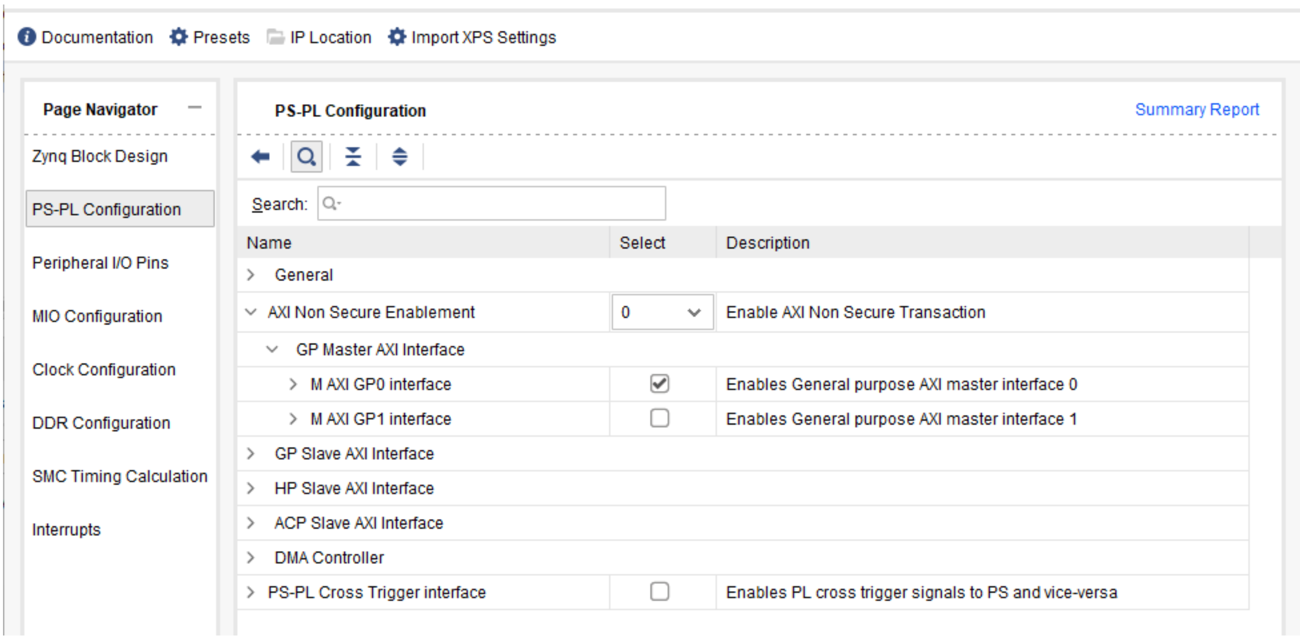
1.2.4 Search 창에 zynq 입력 후 ZYNQ7 Processing System 더블 클릭

1.2.5 Diagram 의 Run Block Automation 클릭 후 OK 클릭

1.2.6 Zynq7 Processing System 더블 클릭 후 MIO Configuration 클릭 후 ENET 0, USB 0, 선택 해제

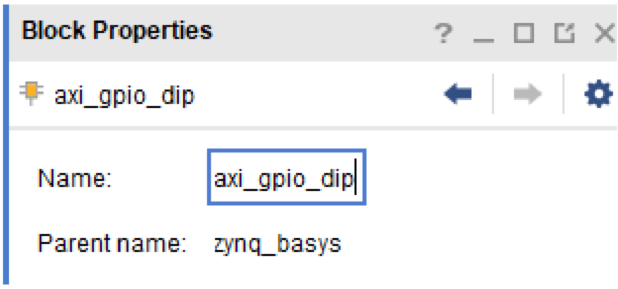


1.2.7 PL-PS Configuration 클릭 후 AXI Non Secure Enablement 확장 후 M AXI GP0 interface 선택 확인 OK 클릭

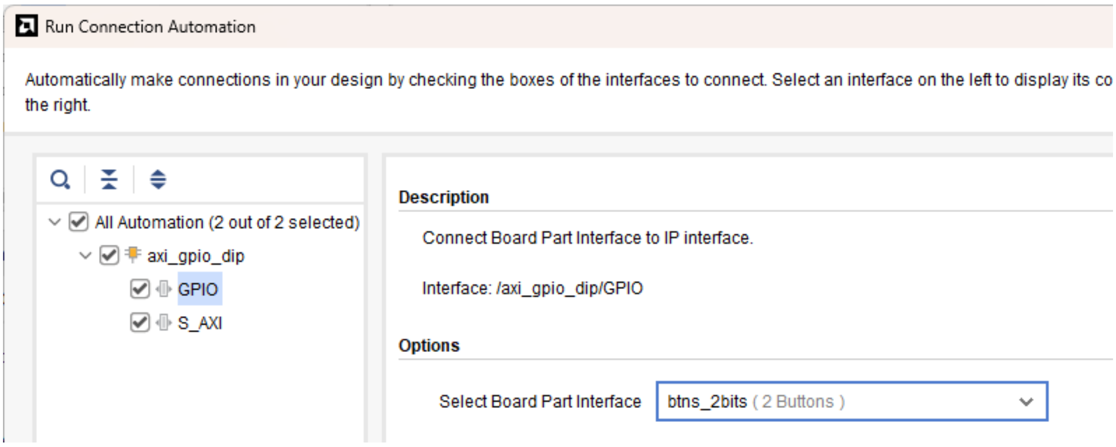


1.2.8 Regenerate Layout 클릭

1.2.9 마우스 오른쪽 버튼 클릭 후 Add IP 클릭 -> GPIO 입력 후 AXI GPIO 더블 클릭 AXI GPIO 클릭 후 오른쪽 Block Properties 의 name 을 axi_gpio_dip을 수정 enter



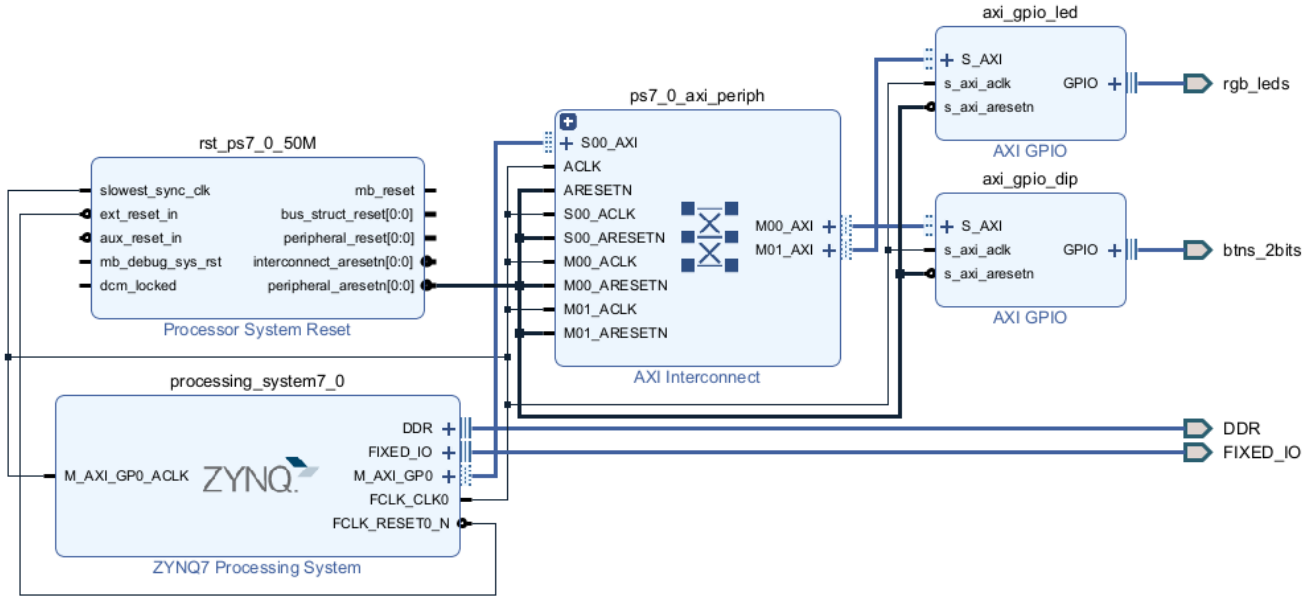
1.2.10 Run Connection Automation 클릭 후 axi_gpio_dip 선택 후 GPIO 클릭하여 Select Board Part interface 에서 btns_2bits(2 buttons) 선택 후 OK클릭



1.2.11 같은 방법으로 GPIO 추가 후 name 을 axi_gpio_led 로 변경

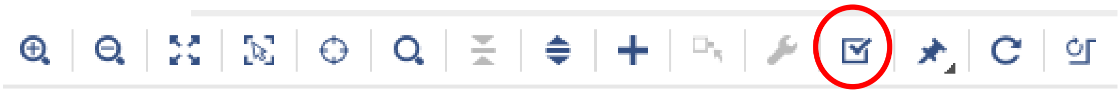
Run connection automation 클릭 후 GPIO를 rgb_leds(2 RGB LEDs) 선택 후 OK

1.2.12 Regenerate layout 클릭 후 완성되 Diagram이 다음과 같은 지 확인



1.2.13 Address Editor 탭 클릭하여 address 확인

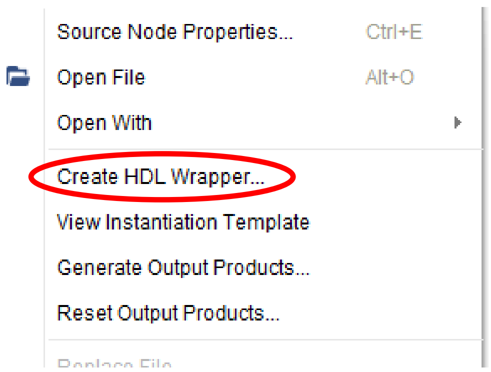
1.2.14 Diagram 탭 클릭 하고 Validate Design(F7) 클릭 하여 error 체크



Warring message OK 클릭

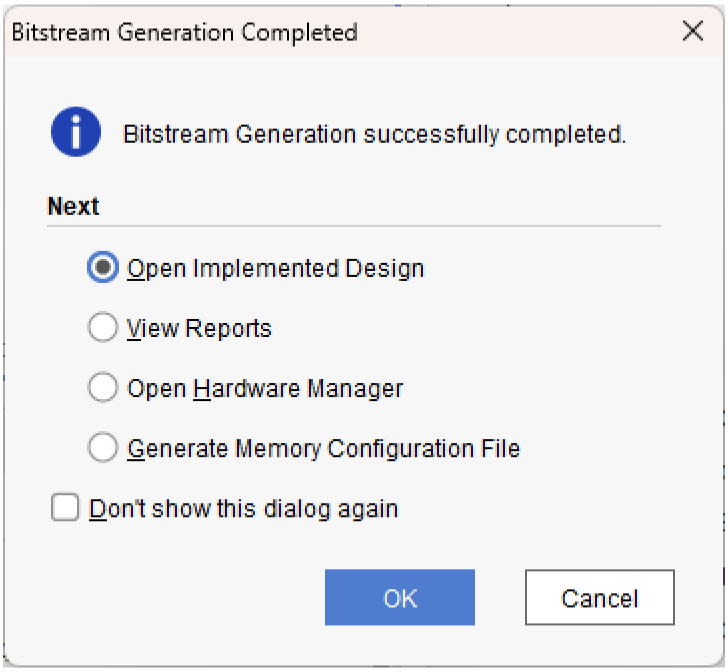
1.2.15 Block design의 x 클릭하여 종료 하면서 저장

1.2.16 Source -> Design Sources -> Zynq_basys(zynq_basys.bd) 선택 후 마우스 오른쪽 버튼 클릭 후 Create HDL Wrapper 클릭 OK

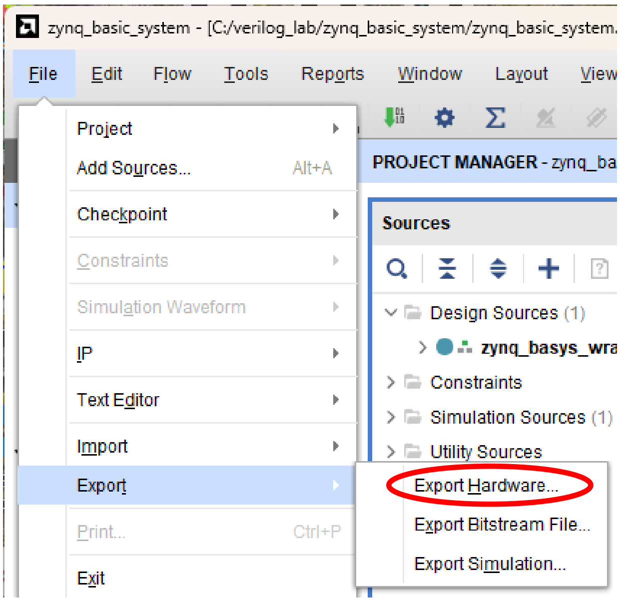


1.2.17 Generate bitstream 클릭 하여 bitstream 생성

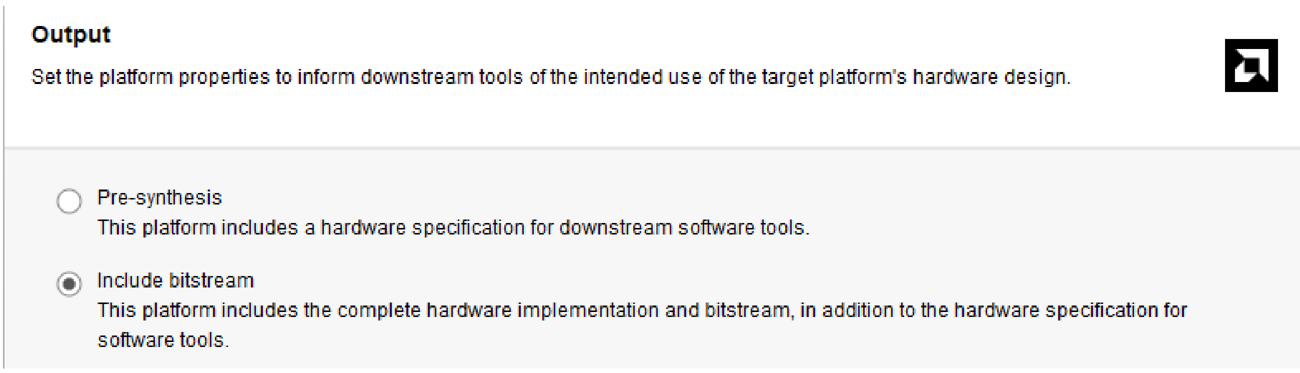
1.2.18 Bitstream 생성 후 cancel 클릭



1.2.19 Filw -> export -> export hardware 클릭



1.2.20 Output 화면에서 include bitstream 선택 후 next



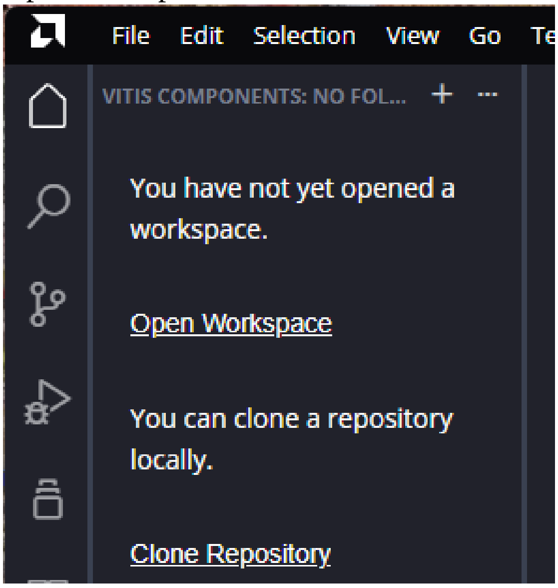
1.2.21 File 화면에서 XSA file name을 zynq_basys로 변경 후 next -> ok

1.2.22 Vivado 종료

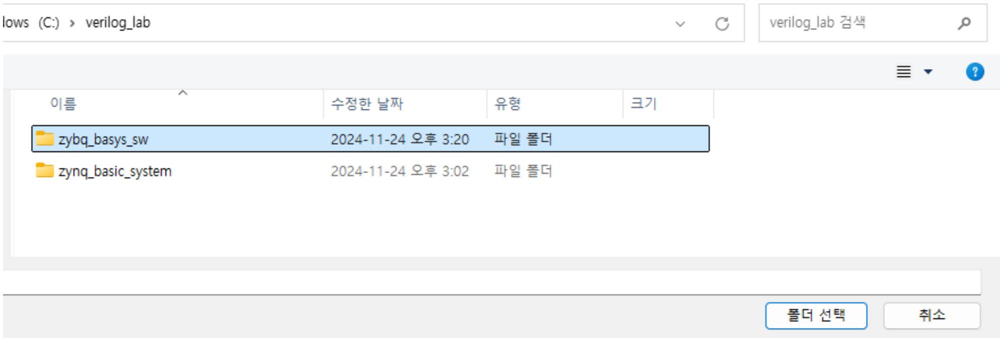
1.3 Vitis 실행

1.3.1 Vitis 실행

Welcome 화면 close
Open workspace 클릭

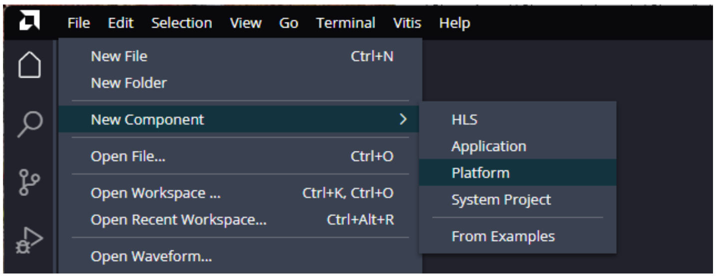


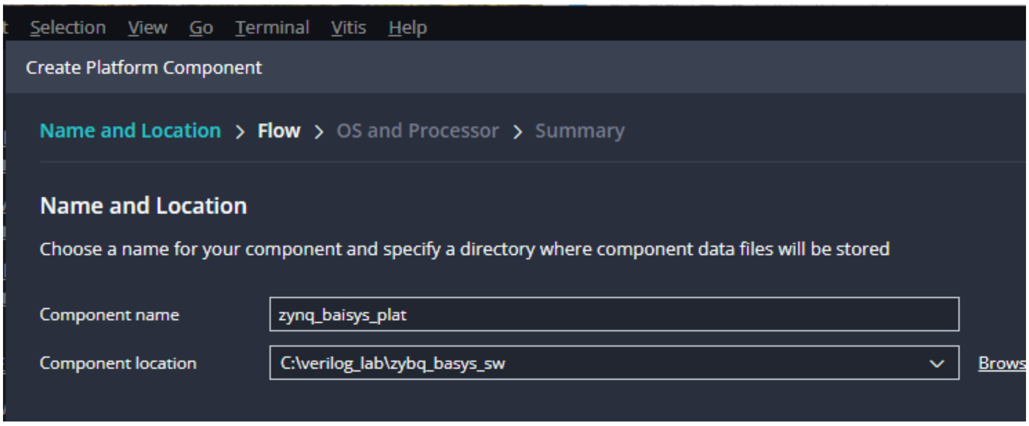
1.3.2 폴더 선택 화면에서 Verilog_lab 선택 후 zynq_basys_sw 폴더 생성 후 zynq_basys_sw 선택



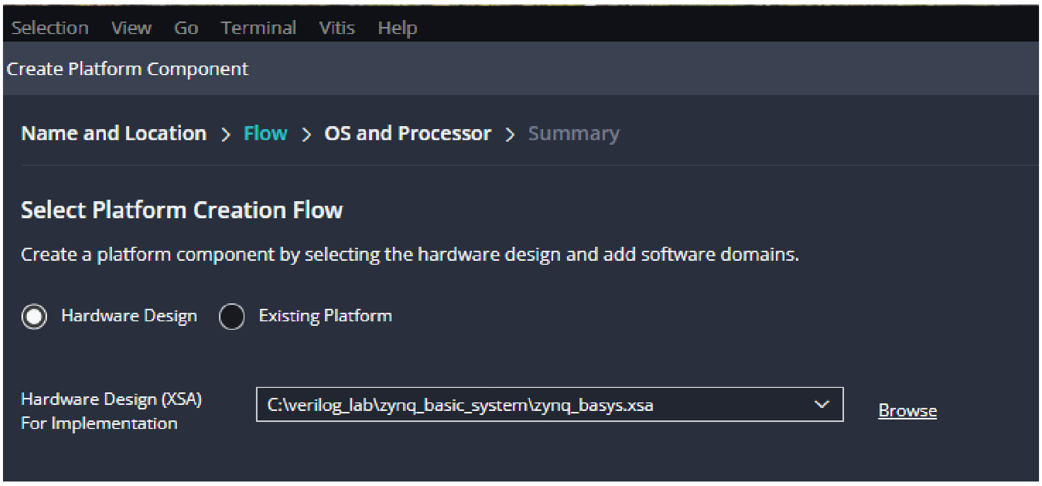
1.3.3 Platform component 생성

File -> New Component -> platform 클릭
Component name 을 zynq_basys_plat로 수정 후 next 클릭





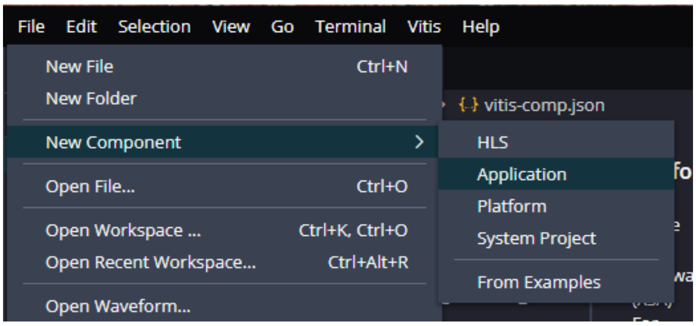
1.3.4 Select platform creation Flow에서 hardware Design(XSA) for implementation 에서 Browse 클릭 후 zynq_basic_system 의 zynq_basys.xsa 선택 후 열기

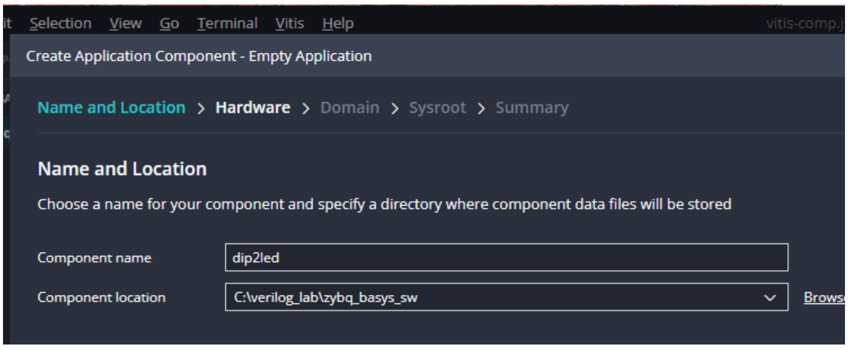


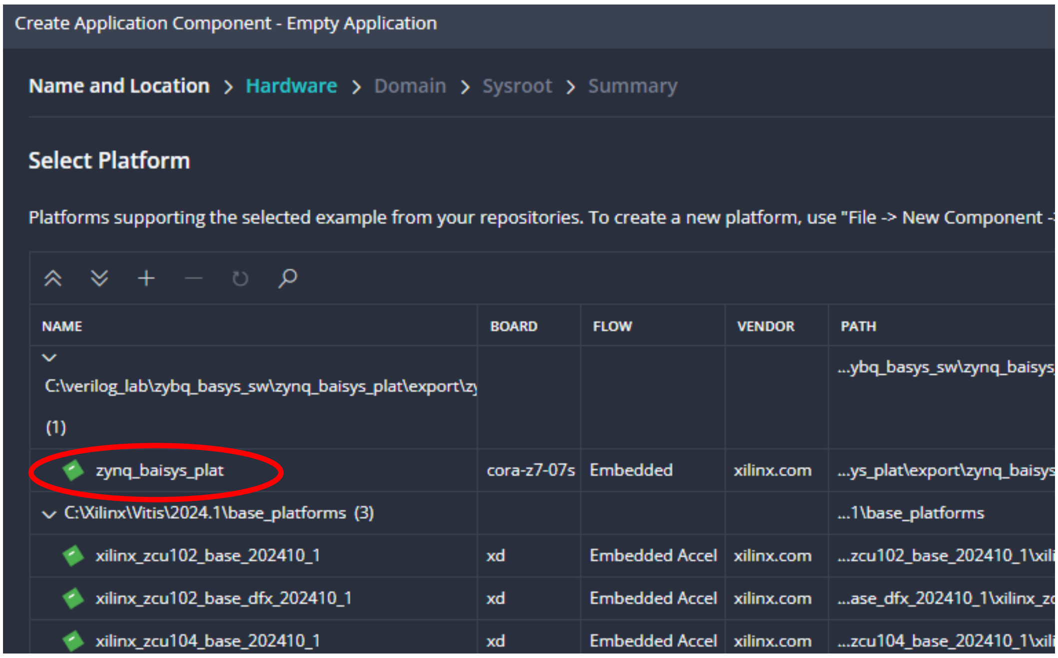
1.3.5 Next -> next -> finish 클릭

1.3.6 Application component 생성

File -> New Component -> Application 클릭
Component name 을 dip2led로 변경 후 next
Select platform 에서 zynq_basys_pat 선택
Next -> next -> finish

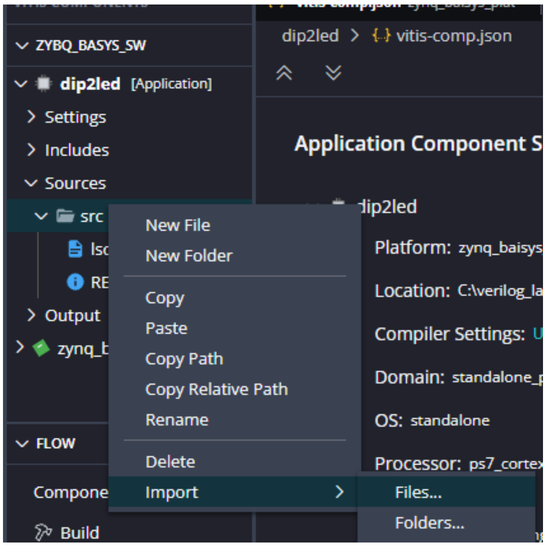






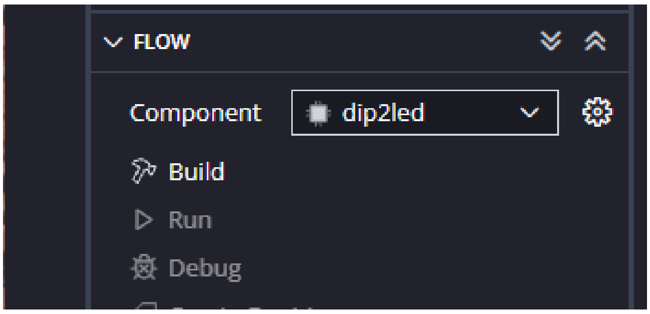
1.3.7 Application source 추가

Dip2led -> source -> src -> import – files 클릭 후 배포한 dip2led.c 선택



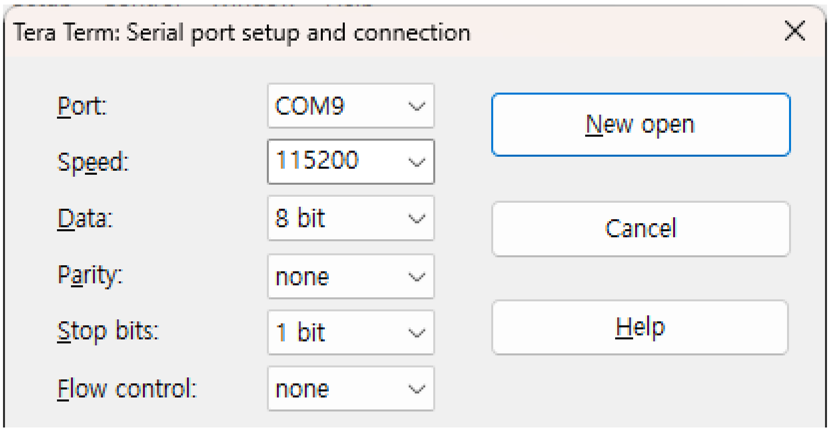
1.3.8 Application build

Component를 dip2led 선택 후 build 클릭



1.3.9 동작 확인

Cora z7 board 연결
Teraterm 실행



1.3.10 Flow 에서 component dip2led 선택 확인 후 run 클릭

Teratem과 보드에서 확인